MIT OpenCourseWare

7.012

Introductory Biology 7.012 — Introductory Biology May 2026 **EXTREMELY**

HARD – MIT LEVEL

Time Limit: 3 hours

Total Marks: 100

1. (4 points) Construct an original proof-level problem involving Biochemistry for 7.012 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

2. (8 points) Prove the fundamental theorem concerning Genetics in the context of 7.012 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

3. (6 points) Analyze the limitations of standard approaches to Molecular Biology when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

4. (3 points) Construct an original proof-level problem involving Cell Biology for 7.012 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

5. (2 points) Analyze the limitations of standard approaches to Development when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

6. (5 points) Prove the fundamental theorem concerning Immunology in the context of 7.012 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

7. (6 points) Prove the fundamental theorem concerning Biochemistry in the context of 7.012 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

8. (6 points) Analyze the limitations of standard approaches to Genetics when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

9. (5 points) Analyze the limitations of standard approaches to Molecular Biology when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

10. (7 points) Analyze the limitations of standard approaches to Cell Biology when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

11. (5 points) Prove the fundamental theorem concerning Development in the context of 7.012 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

12. (4 points) Analyze the limitations of standard approaches to Immunology when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

13. (5 points) Construct an original proof-level problem involving Biochemistry for 7.012 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

14. (3 points) Analyze the limitations of standard approaches to Genetics when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

15. (3 points) Construct an original proof-level problem involving Molecular Biology for 7.012 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

16. (3 points) Prove the fundamental theorem concerning Cell Biology in the context of 7.012 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

17. (3 points) Construct an original proof-level problem involving Development for 7.012 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

18. (5 points) Analyze the limitations of standard approaches to Immunology when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

19. (2 points) Analyze the limitations of standard approaches to Biochemistry when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (2 points) Construct an original proof-level problem involving Genetics for 7.012 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

21. (2 points) Analyze the limitations of standard approaches to Molecular Biology when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

22. (4 points) Construct an original proof-level problem involving Cell Biology for 7.012 (Introductory Biology) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

23. (3 points) Analyze the limitations of standard approaches to Development when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

24. (2 points) Analyze the limitations of standard approaches to Immunology when applied to edge cases in 7.012. Construct explicit counterexamples showing where intuitive reasoning fails.

25. (2 points) Prove the fundamental theorem concerning Biochemistry in the context of 7.012 (Introductory Biology). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

Solution approach: First recall the definitions and key theorems related to Biochemistry from 7.012. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 2:

The proof follows from the definitions and properties established in 7.012. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 3:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Biology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 4:

Solution approach: First recall the definitions and key theorems related to Cell Biology from 7.012. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 5:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Development. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 6:

The proof follows from the definitions and properties established in 7.012. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 7:

The proof follows from the definitions and properties established in 7.012. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Genetics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 9:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Biology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 10:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Cell Biology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 11:
The proof follows from the definitions and properties established in 7.012. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:
The edge case analysis reveals the subtle assumptions underlying standard treatments of Immunology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 13:
Solution approach: First recall the definitions and key theorems related to Biochemistry from 7.012. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 14:
The edge case analysis reveals the subtle assumptions underlying standard treatments of Genetics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 15:
Solution approach: First recall the definitions and key theorems related to Molecular Biology from 7.012. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 16:
The proof follows from the definitions and properties established in 7.012. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 17:
Solution approach: First recall the definitions and key theorems related to Development from 7.012. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 18:
The edge case analysis reveals the subtle assumptions underlying standard treatments of Immunology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 19:
The edge case analysis reveals the subtle assumptions underlying standard treat-

ments of Biochemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

Solution approach: First recall the definitions and key theorems related to Genetics from 7.012. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 21:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Biology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 22:

Solution approach: First recall the definitions and key theorems related to Cell Biology from 7.012. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 23:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Development. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 24:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Immunology. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 25:

The proof follows from the definitions and properties established in 7.012. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.